

Cardiac CINE MRI Semantic Segmentation Utilizing a Deep Learning Approach

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Abstract—Diagnosis of cardiac problems including Myocardial infarction damage/scarring, Cardiomyopathy, and valve diseases heavily relies on modern non-invasive magnetic resonance imagery (MRI). Analysis of CMR images is crucial for the early detection of fatal cardiac diseases. However, this task is human-intensive and requires manual inspection and delineation while being prone to human errors. Machine learning and deep learning approaches proved to offer excellent quality in medical imagery including brain and colon MRI. In this study, we propose the practical study of 5 different deep-learning architectures to perform semantic segmentation of the heart's key organs with 3 different imaging planes, 4CH, 2CH and SAX. Proposing a new ORACM-based architecture proving slight improvement over the state-of-the-art models. Our proposed architectures are trained and tested on 105 real-life cases of manually annotated CMR images resulting in a state-of-the-art dice score of 98% and an F1-score of 99%. Our proposed architecture can be used as a first step analysis before being input for other disease-specific models or simply to aid the medical human expert for easy decision making.

Keywords—Medical diagnostic imaging, Cardiac MRI, Medical Image Processing, Deep Learning, Semantic segmentation, Supervised learning, Hyperparameter optimization, Convolutional neural networks, U-Net, Fully-Convolutional Networks

I. INTRODUCTION

Diagnosing cardiac diseases, scars and abnormalities has been dictated by medical imagery in the past few decades. Computed Tomography (CT) scans was largely used to diagnose coronary artery diseases and for calcium scoring [1], it leverages X-rays with computer-aided reconstruction. CT scans are known for being fast and cheap but excessive radiation acts as its main downside. CT scans underwent major improvements since the utilization of deep learning to reconstruct x-ray images [2]. Echocardiography is another imaging technique that is used to assess dynamic functions of the heart by sending sound waves to create real-time moving images of the heart during blood flow. Echocardiography is used in detecting early signs of Heart Failures (HF) and Coronary Artery Diseases (CAD) as well as diagnosing patients with diabetes mellitus [3]. However, Echocardiography yields very low resolution images/frames that fail to discriminate core tissue of the heart for evaluating volume or density or area of the heart's ventricles or the myocardium or the volume of cavity of the chambers. Magnetic Resonance Imaging (MRI)

was first introduced in Australia and used on a live human patient on July 3, 1977, by Dr. Raymond Damadian and his team, who scanned a man's chest to successfully demonstrate the technology. It is a non-invasive and safe technique that can capture the core tissues of the heart for medical diagnosis of several diseases and for detecting scars and to measure the amount of cardiac fatty tissues around the heart [4]. However, MRI is slow, expensive and needs manual delineation and extensive manual efforts to segment the heart's core tissue areas which is prone to human errors and inaccuracies as well as being expensive for this human labor hours. In this work, we propose multiple deep learning based approaches to overcome the issue of manual delineation and human annotation of CMR images and also compare the results between our approaches in-relation to the related work in literature. We also present a pipeline of preprocessing the raw CMR images to make them suitable for computational tasks and prevent unpredicted results while boosting explainability of our approach. In addition to proposing the usage of Online Region-Based Active Contour Model (ORACM) jointed with a deep learning model (FCN). The rest of the paper is organised as follows: We present the related work in Section II, we present the different proposed architectures in section III. A custom loss function is stated in IV, and we explore the available dataset in V. Results are studied in section VI and finally concluding our work in section VII.

II. RELATED WORK

Cardiovascular diseases remain among the leading causes of mortality worldwide, making early diagnosis and accurate cardiac assessment essential in modern healthcare systems. Cardiac Magnetic Resonance Imaging (CMR) is considered one of the most reliable imaging modalities for evaluating cardiac anatomy, ventricular function, myocardial abnormalities, tumors, and structural heart diseases because of its high spatial resolution and superior soft-tissue contrast [19]. However, manual segmentation of cardiac structures from cine MRI images is time-consuming, labor-intensive, and highly dependent on clinical expertise. Therefore, deep learning-based automatic cardiac MRI segmentation methods have received significant attention in recent years.

TABLE I
SUMMARY OF RELATED WORK

Paper	Technique	Findings	Limitations	Research Gaps
Gradient-Based Learning Applied to Document Recognition [5]	CNN	Effective hierarchical feature extraction and automatic image representation learning	Limited pixel-wise segmentation capability	Need for accurate medical image segmentation frameworks
Fully Convolutional Networks for Semantic Segmentation [6]	FCN	Enabled end-to-end semantic segmentation	Coarse segmentation boundaries due to repeated pooling	Preservation of fine anatomical cardiac structures
U-Net: Convolutional Networks for Biomedical Image Segmentation [7]	U-Net	Improved biomedical image segmentation accuracy using encoder-decoder architecture	Sensitive to data variability and noise	Limited multi-view cardiac MRI evaluation
UNet++: A Nested U-Net Architecture for Medical Image Segmentation [8]	UNet++	Enhanced multi-scale feature fusion and segmentation accuracy	Increased computational complexity	Limited application in multi-view cine MRI analysis
Attention U-Net: Learning Where to Look for the Pancreas [9]	Attention U-Net	Improved focus on important anatomical regions	Increased computational cost	Limited stability analysis across cardiac MRI views
Recurrent Residual Convolutional Neural Network Based on U-Net (R2U-Net) [10]	R2U-Net	Improved feature propagation and segmentation robustness	High memory consumption	Scalability challenges on large medical datasets
An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale [11]	Vision Transformer (ViT)	Captured global contextual image dependencies effectively	Requires large-scale datasets and high computation	Limited direct medical segmentation applications
TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation [12]	TransUNet	Combined CNN local features with transformer global representation	High GPU memory usage	Need for lightweight implementations in medical imaging
UNETR: Transformers for 3D Medical Image Segmentation [13]	UNETR	Strong transformer-based segmentation performance	Complex training process	Limited comparative analysis with CNN and FCN methods
Swin-Unet: Unet-Like Pure Transformer for Medical Image Segmentation [14]	Swin-Unet	Improved multi-scale contextual representation	High model complexity	Limited real-world deployment in cardiac MRI systems
Segment Anything [15]	SAM	Strong segmentation generalization capability	Requires extensive computational resources	Limited adaptation to cardiac cine MRI segmentation
MedSAM: Segment Anything in Medical Images [16]	Foundation Model	Improved medical image segmentation generalization	Requires extensive fine-tuning	Limited validation on cine cardiac MRI datasets
Automated Cardiovascular Magnetic Resonance Image Analysis with Fully Convolutional Networks [17]	FCN-Based Cardiac MRI Segmentation	Automated cardiac chamber segmentation and functional analysis	Limited multi-view evaluation	Insufficient analysis across SAX, 2CH, and 4CH views
Convolutional Neural Networks with Shape Priors for Cardiac MRI Segmentation [18]	CNN + Shape Priors	Improved anatomical consistency in cardiac segmentation	Sensitive to image variability	Limited generalization across datasets
Statistical Normalization Techniques for Magnetic Resonance Imaging [19]	MRI Normalization	Improved MRI consistency and preprocessing quality	Dataset dependency	Need for unified normalization frameworks
A Comparative Analysis of Data Augmentation Approaches for MRI Scan Images [20]	MRI Data Augmentation	Improved robustness and generalization performance	Augmentation sensitivity	Limited cardiac MRI augmentation studies
USE-Evaluator: Performance Metrics for Medical Image Segmentation Models [21]	Segmentation Evaluation Metrics	Improved reliability of segmentation evaluation	Complex evaluation process	Need for unified cardiac MRI evaluation metrics
Proposed Work	CNN + FCN + U-Net	Comparative multi-view cardiac MRI segmentation across SAX, 2CH, and 4CH views	Limited to supervised learning approaches	Future integration with transformer-based hybrid frameworks

Early deep learning approaches for medical image analysis mainly relied on Convolutional Neural Networks (CNNs). LeCun et al. [5] demonstrated the effectiveness of CNN architectures in hierarchical feature extraction and automatic image representation learning. Although CNN-based approaches achieved strong performance in image classification tasks, they were not originally designed for dense pixel-wise medical image segmentation, which limited their ability to accurately preserve anatomical boundaries in cardiac MRI applications.

To address these limitations, Long et al. [6] introduced Fully Convolutional Networks (FCNs), replacing fully connected layers with convolutional operations to enable end-to-end semantic segmentation. FCNs significantly improved dense prediction and spatial segmentation performance; however,

repeated pooling and downsampling operations often resulted in coarse segmentation boundaries and loss of fine anatomical details, which are critical in cardiac MRI analysis.

A major advancement in biomedical image segmentation was achieved through the introduction of U-Net by Ronneberger et al. [7]. The U-Net architecture employed an encoder-decoder framework with skip connections that transferred high-resolution spatial features from the encoder to the decoder. This architecture substantially improved localization accuracy and became one of the most widely adopted segmentation models in medical imaging and cardiac MRI analysis.

Several extensions of U-Net were later proposed to further improve segmentation performance. Zhou et al. [8] introduced UNet++, which utilized nested dense skip pathways to enhance multi-scale feature fusion and segmentation accuracy. Oktay

et al. [9] proposed Attention U-Net, integrating attention mechanisms to improve focus on important anatomical structures while suppressing irrelevant background information. Similarly, Alom et al. [10] developed R2U-Net by combining recurrent convolutional layers with residual learning to improve feature propagation and segmentation robustness.

Transformer-based architectures have also demonstrated strong performance in computer vision and medical image analysis. Dosovitskiy et al. [11] introduced the Vision Transformer (ViT), showing that self-attention mechanisms can effectively capture long-range dependencies within images. Building upon this concept, Chen et al. [12] proposed TransUNet, combining transformer encoders with CNN-based decoders to extract both global contextual information and local spatial features. Hatamizadeh et al. [13] further introduced UNETR, a transformer-driven segmentation framework for medical imaging applications. More recently, Cao et al. [14] proposed Swin-Unet, which utilized shifted-window transformer mechanisms to improve multi-scale contextual representation in medical image segmentation tasks.

Foundation models have recently emerged as promising approaches in medical image analysis. Kirillov et al. [15] introduced the Segment Anything Model (SAM), which demonstrated strong generalization capability across various segmentation tasks. Based on this concept, MedSAM [16] adapted SAM for medical image segmentation applications and achieved promising performance across multiple medical imaging modalities. Nevertheless, these approaches generally require substantial computational resources and extensive fine-tuning for cardiac cine MRI applications.

In cardiac MRI specifically, Bai et al. [17] proposed an FCN-based framework for automated cardiovascular magnetic resonance analysis, enabling automatic segmentation of cardiac chambers and functional assessment. Zotti et al. [18] further improved anatomical consistency by incorporating shape priors into CNN-based cardiac MRI segmentation models.

Several studies also emphasized the importance of pre-processing and normalization techniques in MRI-based deep learning systems. Shinohara et al. [19] demonstrated that MRI normalization significantly improves image consistency and segmentation performance across datasets. Safdar et al. [20] showed that data augmentation techniques can improve generalization capability and robustness in MRI image analysis. Furthermore, Ostmeier et al. [21] highlighted the importance of reliable evaluation metrics for medical image segmentation, especially in challenging annotation settings.

Although significant progress has been achieved in deep learning-based cardiac MRI segmentation, several challenges still remain. Previous studies have investigated different segmentation architectures, including CNN-, FCN-, and U-Net-based models, and some works reported comparative evaluations among these approaches. However, many existing studies focused primarily on improving segmentation accuracy on specific datasets or single cardiac MRI views, while limited attention was given to establishing a unified comparative framework across multiple cine MRI representations such as

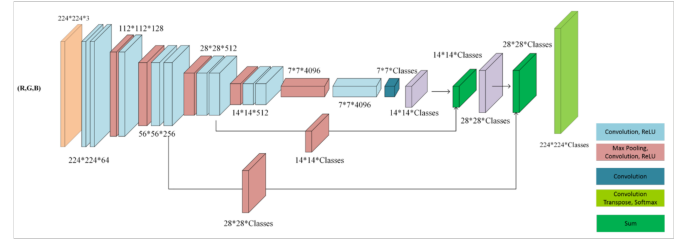


Fig. 1. FCN with chosen skip-connections architecture

SAX, 2CH, and 4CH views. In addition, several approaches emphasized architectural complexity and performance improvements without sufficiently analyzing training consistency, model robustness, and practical applicability in multi-view cardiac MRI segmentation tasks.

Motivated by these limitations, the present work introduces a comparative multi-view cardiac MRI segmentation framework based on CNN, FCN, and U-Net architectures. Unlike previous studies that focused on a single architecture or individual MRI representation, the proposed framework evaluates segmentation performance consistently across SAX, 2CH, and 4CH cine MRI views using unified preprocessing, training, and evaluation procedures. The proposed study aims to provide a balanced comparative analysis of segmentation accuracy, robustness, and practical applicability for automated cardiac MRI segmentation.

III. METHODOLOGY

A. FCN with Skip Connections Architecture

Since our main goal in this problem is not to create images as in generative AI, we want to make use of the original image in order to annotate some pixels inside it. Preserving the features from the original image is crucial as we are not creating a whole new non-existing image. We implemented some skip-connections where we take a copy of the image after some certain layers in the encoder and simply concatenate this image to its counterpart in the decoder hence, preserving some of the original image features to be utilized during the up-convolution phases. Those skip-connections are not totally from each encoder phase to each decoder phase as the U-Net's architecture, but from some chosen layers in the encoder to some chosen layers in the decoder. The general architecture of this approach is shown in figure 1. The update is that we implemented 2 skip-connections from the third and fourth stage in the encoder, to the second and third stage in the decoder respectively, using a summation operation since both images (skipped from the encoder to its counterpart in the decoder) have the same dimensions. Training this type of network took longer time and required more processing resources (GPU, RAM and CPU utilization). However, its results were already explainable and robust.

B. Edge-Guided FCN (EG-FCN)

Building upon the work of [22] which proposed edge extraction and concatenating edges to the original image before

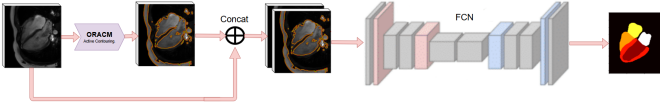


Fig. 2. Edge-Guided FCN using ORACM active contouring

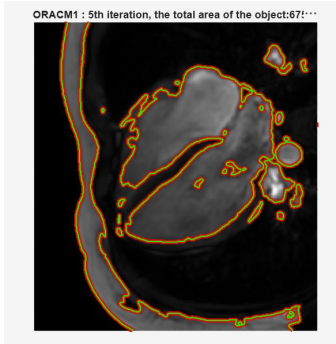


Fig. 3. Applying ORACM active contouring on a sample 4CH test case

inputting both into deep learning models for faster and more accurate training. We proposed a new architecture utilizing Active Contours using M. Fatih’s ORACM [23] jointed with the FCN. Our proposed architecture works by extracting the contours of the image, concatenating the contours to the actual image as a second channel, then input the new image to the FCN for training. Same pipeline is used for testing. The general architecture is shown in figure 2. Showing a slight jump in segmentation performance without improving the training speed by much. The output of a sample 4CH after applying ORACM on it is shown in figure 3

C. U-Net Architecture

To achieve accurate and reliable cardiac MRI segmentation, a U-Net–based deep learning architecture was implemented in this study due to its effectiveness in biomedical image segmentation tasks and its ability to preserve both semantic and spatial information [7]. The adopted model was designed to segment cardiac cine MRI images obtained from multiple cardiac views, including Short-Axis (SAX), Two-Chamber (2CH), and Four-Chamber (4CH) representations.

Figure 4 presents the detailed architecture of the implemented U-Net framework used in this work.

The implemented U-Net follows a symmetric encoder–decoder architecture composed of three major components: the encoder (contracting path), the bottleneck layer, and the decoder (expanding path). The overall structure was specifically designed to balance contextual feature extraction and accurate pixel-level localization, which are essential requirements in cardiac MRI segmentation tasks.

The input to the network consists of grayscale cardiac MRI slices resized to 256×256 pixels. Prior to training, all MRI images were normalized to improve intensity consistency across different scans and patients. This preprocessing step reduces variability caused by imaging conditions and enhances training stability.

The encoder path is responsible for extracting hierarchical spatial and semantic features from the input MRI images. Each encoder level contains two consecutive convolutional layers with kernel size 3×3 , followed by Rectified Linear Unit (ReLU) activation functions. Padding was applied to preserve spatial dimensions after convolution operations. A 2×2 max-pooling layer with stride 2 was then used to progressively reduce spatial resolution while increasing the receptive field of the network.

As illustrated in Fig. 4, the number of feature channels increased gradually through the encoder levels from 64 to 128, 256, and 512 feature maps. This progressive increase enables the model to learn low-level image characteristics such as edges and textures in earlier layers while capturing more complex anatomical structures and contextual cardiac information in deeper layers.

At the deepest stage of the network, referred to as the bottleneck layer, the extracted feature maps reached a spatial size of 16×16 with 1024 feature channels. This layer provides a compact semantic representation of the cardiac structures and surrounding tissues. The bottleneck stage plays an important role in capturing high-level contextual information required for accurate segmentation of ventricular cavities, myocardium, and cardiac boundaries.

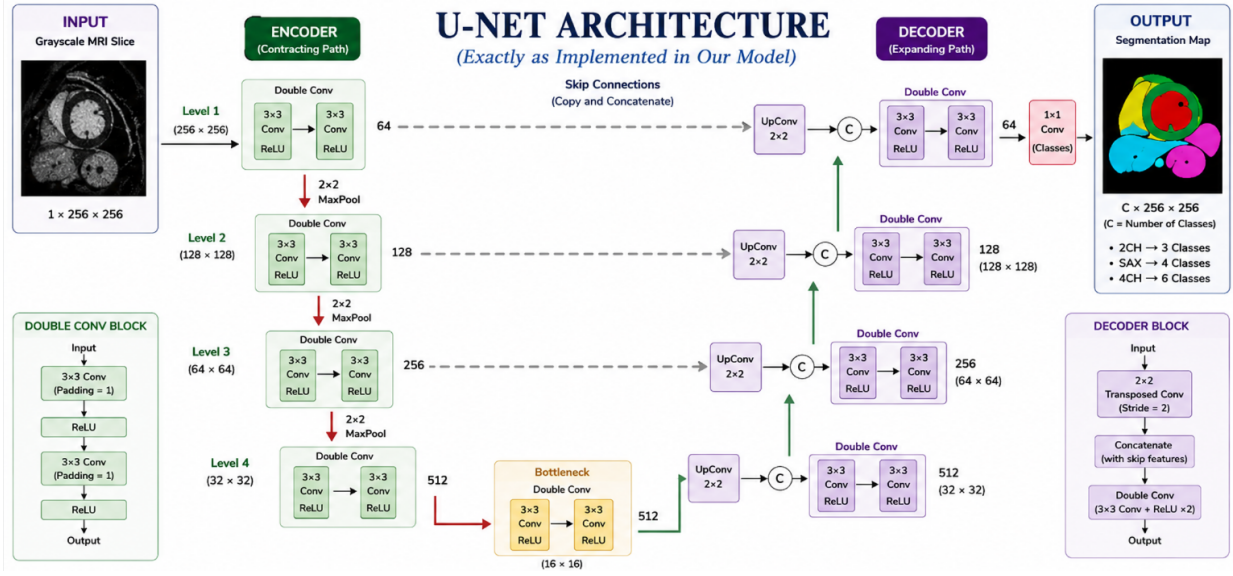
The decoder path reconstructs the segmentation mask by progressively restoring the spatial resolution of the feature maps. Each decoder level begins with a 2×2 transposed convolution (up-convolution) operation used for upsampling the feature maps. The upsampled features are then concatenated with the corresponding encoder features through skip connections.

The skip connections represent one of the most important components of the U-Net architecture. During the encoder downsampling process, some fine spatial details may be lost due to repeated pooling operations. Therefore, skip connections directly transfer high-resolution encoder features to the decoder layers, allowing the network to recover anatomical boundaries and preserve fine structural information. This mechanism significantly improves localization accuracy and enhances segmentation quality, particularly in medical imaging applications where precise boundary delineation is critical.

After concatenation, two additional 3×3 convolutional layers followed by ReLU activation functions are applied to refine the reconstructed feature maps. The decoder gradually restores the original spatial resolution while reducing the number of feature channels until the final segmentation map is generated.

At the final stage of the network, a 1×1 convolutional layer is used to project the extracted features into pixel-wise class probabilities corresponding to the target cardiac structures. The output segmentation mask preserves the same spatial dimensions as the original input image.

The implemented model was trained using the Adam optimization algorithm with a learning rate of 10^{-4} . Cross-Entropy Loss was employed as the primary optimization objective for pixel-wise classification. In addition, mixed



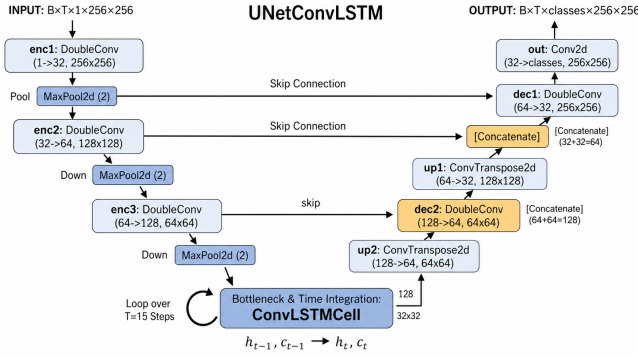


Fig. 6. Overview of the proposed UNet-ConvLSTM architecture. Consecutive cine cardiac MRI frames are first processed by a convolutional encoder to extract hierarchical spatial features. A ConvLSTM bottleneck captures temporal dependencies between successive frames, while skip connections transfer high-resolution spatial information to the decoder for accurate segmentation map reconstruction.

six classes.

For training, we used a hybrid loss function: half weighted cross-entropy and half Dice loss. We did this because medical imaging datasets usually have class imbalance, but we still wanted to keep the boundaries sharp.

E. UNet-ConvLSTM Architecture

Cardiac cine MRI consists of a sequence of frames that capture the heart throughout the cardiac cycle. Unlike static medical images, neighboring frames are strongly correlated, as they represent different phases of the same cardiac motion. Therefore, an effective segmentation model should be capable of learning not only spatial anatomical features from individual frames but also temporal relationships across the sequence.

To address this challenge, a hybrid UNet-ConvLSTM architecture was employed, combining the strong feature extraction capability of U-Net with the temporal modeling ability of ConvLSTM. The overall architecture is illustrated in Fig. 6.

The encoder consists of three convolutional stages. Each stage contains two consecutive convolutional layers followed by batch normalization and ReLU activation. As the input progresses through the encoder, max-pooling operations gradually reduce the spatial resolution while increasing the number of feature channels. This hierarchical representation enables the network to capture both low-level image details and high-level semantic information describing cardiac anatomy.

At the bottleneck, a ConvLSTM cell is introduced to model temporal dependencies within the cine MRI sequence. Unlike conventional U-Net architectures that process frames independently, the ConvLSTM maintains hidden and cell states across time, allowing information from previous frames to influence the representation of the current frame. This recurrent mechanism enables the network to capture cardiac motion patterns and preserve temporal consistency throughout the sequence.

The decoder reconstructs the segmentation maps through a series of transposed convolution layers that progressively restore the spatial resolution. To recover fine anatomical details

lost during downsampling, skip connections are established between corresponding encoder and decoder stages. These connections provide direct access to high-resolution spatial features and improve the localization of anatomical boundaries.

At each decoding stage, the upsampled features are concatenated with the corresponding encoder features and further refined using convolutional blocks. This feature fusion strategy combines contextual information learned at deeper layers with detailed spatial information from earlier layers, leading to more discriminative feature representations.

Finally, a 1×1 convolutional layer projects the decoder output into the predefined segmentation classes, producing a pixel-wise prediction for each frame in the sequence. By integrating spatial feature extraction and temporal sequence modeling within a unified framework, the proposed UNet-ConvLSTM architecture is well suited for cine cardiac MRI segmentation and enables robust delineation of cardiac structures across time.

IV. DESIGNING THE LOSS FUNCTION

Since the dice loss itself does not have the ability to differentiate or assign weights or penalties for different classes, and since the images were subject to some misses in the background. We intended to penalize the model when it wrongly masked pixels that originally belong to certain classes. Cross Entropy loss function has this feature of custom loss weights. Hence, we utilized a combined loss function that is a summation of both the Dice loss and the weighted Cross Entropy where we can freely control the class loss weights to penalize the model according to our needs.

Combined Dice Loss and Weighted Cross-Entropy Loss

Softmax Probabilities:

$$p_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=1}^C e^{z_{i,k}}} \quad (1)$$

Weighted Cross-Entropy Loss (per pixel):

$$\mathcal{L}_{WCE} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log(p_{i,y_i}) \quad (2)$$

Dice Loss (excluding background class $c = 0$):

$$\mathcal{L}_{Dice} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{1}{C-1} \sum_{c=1}^{C-1} \frac{2 \sum_{j=1}^{HW} p_{i,c,j} y_{i,c,j} + \epsilon}{\sum_{j=1}^{HW} p_{i,c,j} + \sum_{j=1}^{HW} y_{i,c,j} + \epsilon} \quad (3)$$

Combined Loss:

$$\mathcal{L}_{total} = \mathcal{L}_{WCE} + \mathcal{L}_{Dice} \quad (4)$$

Definitions:

- N : number of samples (batch size)
- C : number of classes
- H, W : spatial dimensions
- w_c : weight for class c
- y_i : ground-truth class label at pixel i
- $y_{i,c,j}$: one-hot encoded ground truth

- $p_{i,c,j}$: predicted probability
- ϵ : smoothing constant

V. DATASET PREPARATION

The dataset is collected and carefully annotated by cardiologists and radiologists from Beijing Anzhen Hospital & collaborative clinical centers[24]. The dataset consists of two collections of images, CINE and LGE, each collection for a special type of problem, the LGE type of imagery is out of scope of this work. The CINE image collection consists of 105 cases -patients- with 3 views per patient which are 4 chamber view (4CH), 2 chamber view (2CH) and short axis view (SAX). Each case has its own annotation files. Each file comes in its original NIfTI format which is the standard image format for MRI in the medical industry, dealing with this type of images programmatically was a challenge. Other challenges will be addressed, as treating NIfTI images as if they were pure PNG or bitmap files might yield biased results.

A. Data Augmentation

Since the provided dataset is not enough for computationally-intensive deep learning models to achieve remarkable results, we studied the use of augmented data and annotate it automatically using another model. However, the usage of augmented data should be minimal compared to the size of the original dataset so as not to introduce artificial artifacts in the new combined dataset and affect the overall performance of our model. Meta AI introduced the Segment Anything Model (SAM) in 2023 [25] which is a pretrained semantic segmentation model that is strengthened by the prompting capabilities to improve the localization of pixels, it is widely used in dynamic MRI and medical imagery for its quick results and ease of use. We propose the usage of SAM to annotate an externally collected small set of CINE MRI images from <https://radiopaedia.org/> to aid the training of our main deep learning model. The portion of augmented data comprises 10% of all used data before splitting into training, testing and validation. The usage of SAM stemmed from its support of prompt engine where we can manually include or exclude parts of each area belonging to a certain class unlike the YOLOv8-dependent approach by Muhammad Farhan et al. in their review of data augmentation [26] which was limited to the capabilities of the pretrained YOLOv8 model to segment parts inside MRI images without being fine-tuned with this type of imagery. 7. This approach however was not fully adopted in our work due to the need for manual point prompts whose automation is outside the scope of this work.

B. Dataset Exploratory Analysis

Since the NIfTI format should not be treated like the average png or bitmaps, each pixel's value range should be inferred from different images, it is found that each NIfTI image has a pixel value ranging from 0 to thousands of arbitrary signal strength units where the intensity for bones is greater than the intensity of fat, and the intensity of fat is greater than the intensity of muscles. Sample histograms of randomly selected

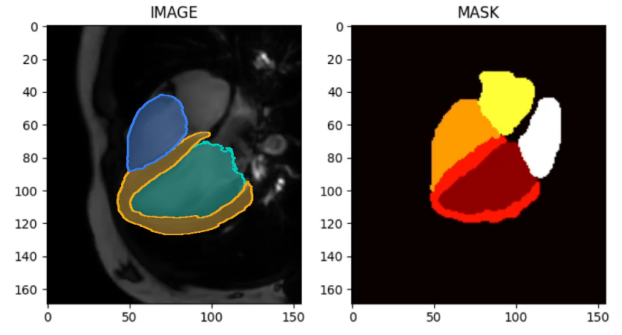


Fig. 7. Using SAM2 during Data Augmentation

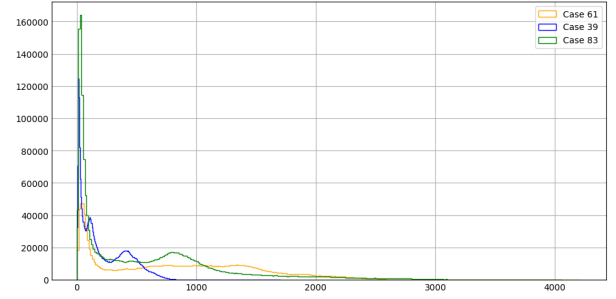


Fig. 8. Different MRI cameras output different histograms with different ranges of values

CMR images from the dataset is shown in figure 8. It is also clear that the max value of images is not 255 nor 65535 nor other max value that is compatible with digital 64-bit storage matrices. We determined that the max value in CMR images follow a certain distribution shown in figure 9. it is found that the max values range from 601 to 7428 acting as a challenge when we input those values into a deep-learning model for training and inference. Those large values may break the model and assign unreal priority to those high value pixels than pixels from other images although both pixels govern the same type of organic tissue.

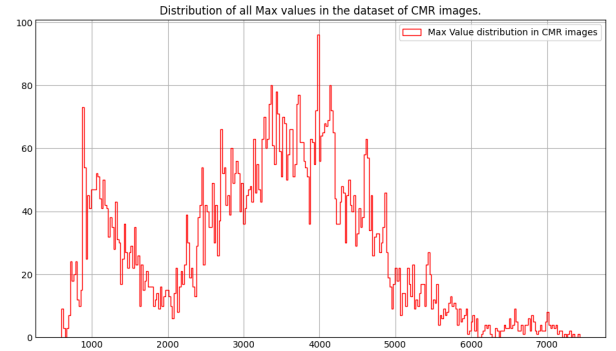


Fig. 9. Distribution of all Max values in CMR images in the dataset.

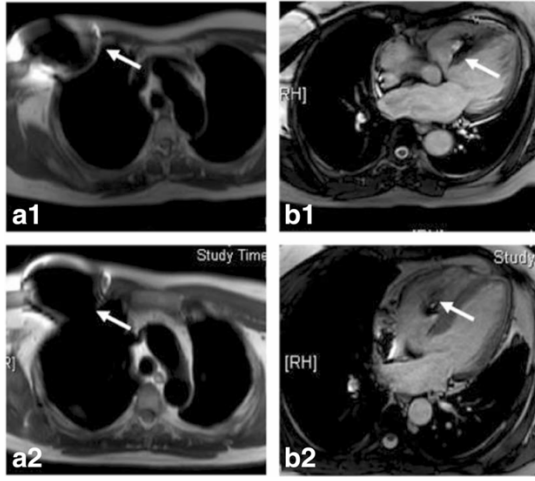


Fig. 10. Very dark spots in CMR due to installation of a pace maker

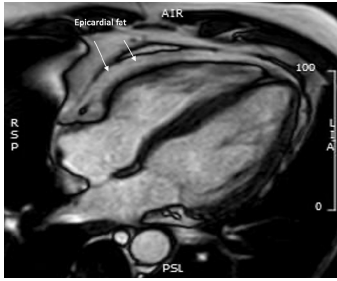


Fig. 11. Fat tissues and excessive water resulting in bright areas in CMR.

C. Scaling Values

Values should be scaled to prevent breaking any statistical model. Hence, we tried different types of scalers which are 1- MinMax scaler, 2- Z-Score Scaler and 3- RobustScaler. The MinMax scaler is a simple normalizer from $[ABS_MIN - ABS_MAX]$ to $[0 - 1]$ but requires the looping on all images to infer the absolute maximum and absolute minimum pixel values. The MinMax scaler is proven to be the most reliable and robust to high variant max values as discussed previously and the scaling is still relative meaning that each part preserves its relative brightness with other regions in the image. After deep analysis, we resorted to the adoption of the local min-max scaler. This is because an outlier image may be input from another camera vendor that has a max value bigger than our computed global ABS_MAX value and such image values will break our model during inference or even during training a global min-max scaler. Side cases can be introduced as stated in [27] such as a case when there is a small piece of metal in the MRI image causing very dark spots or silicon implants causing very bright spots as shown in figures 10 and 11.

Generally speaking, using the local min-max scaler is more reliable in normal cases than the global min-max scaler, however if a certain anomalous case as a case containing a metal piece, this will cause the entire classification of this anomalous case to be affected negatively. It is a trade off and

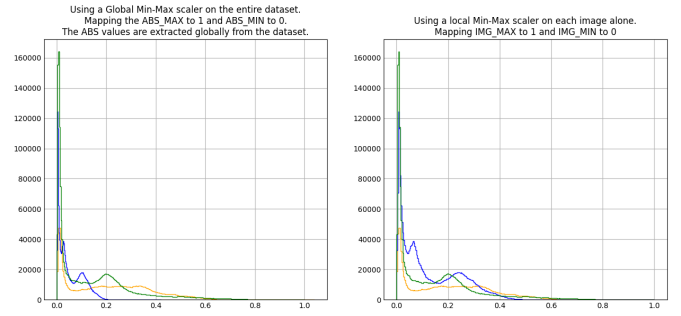


Fig. 12. Global vs Local MinMax scalers and showing histogram mismatch between different MRI cameras

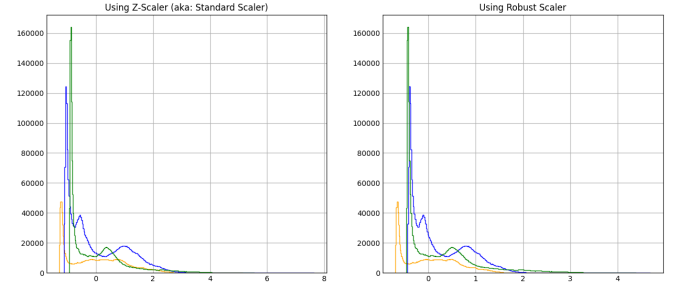


Fig. 13. Comparison between the usage of Robust Scaler and Standard z-Scaler still showing the problem of histogram mismatches.

we will accept the risk of misclassify a single anomalous case than to poison the scaling of our entire dataset from individual anomalous cases. However, the problem of histogram mismatch between MRI cameras is still present and clearly shown in figure 12.

Other scalers were considered such as the Standard -ZScore-Scaler and Robust Scaler as used in Brain MRI normalization in Shinohara et al. study [28]. However, this approach seems not suitable for CMR images, a quick comparison is shown in figure 13 which proves that Z-scaler and robust scaling images of the dataset can bring histogram mismatches and did not correctly solve the problem of histogram mismatches.

Another thing should be considered in minds is that when we use a min-max scaler, due to the very large value of the max value pixel, some features recorded in the early tail of the histogram gets squeezed together after scaling, features became very close to each other to preserve relative distance between the small values to the big values. To address this issue, clipping is considered and tested. By clipping the upper 99 percentile of each image to deter any outliers then perform scaling, this experiment's results is recorded as per figure 14. The values seems fine if we study the histograms, ranges from 0 to 1 is optimal for dealing with mass images, the histograms between different camera vendors seems like they are aligned now with any abnormalities normalized due to the clipping operation.

Final human checking of those images after clipping and after MinMax scaling shows a great shift in the coloring scheme of the MRIs which can alter the human detection/diagnosis of

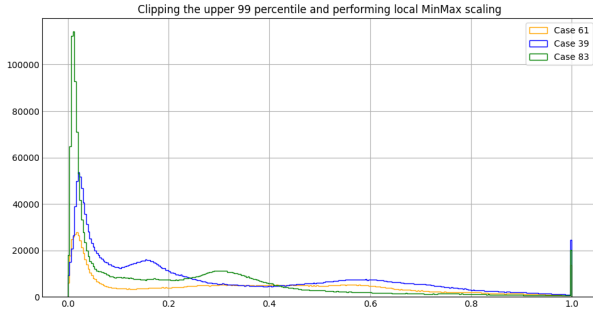


Fig. 14. Clipping 99 percentile and MinMax scaling.

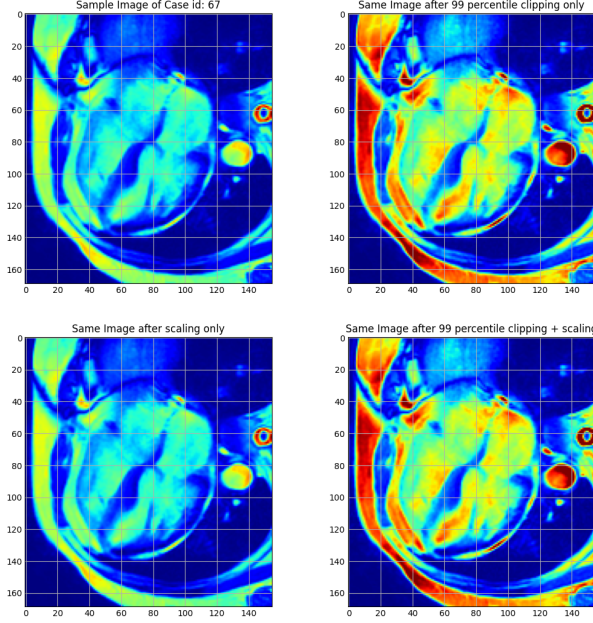


Fig. 15. Comparing the effect of 99% Clipping and MinMax scaling and mixing both operations together

images, this great alteration shown in figure 15 can perplex humans and hence can perplex statistical deep learning models from grabbing the features that humans look for during their daily checkup. The visualization color map was set to "jet" as it captures the slight differences of pixel values in an excellent manner better than the grey-scale visualization to the naked eye.

VI. RESULTS AND EVALUATION

A. Measurement Metrics

Semantic segmentation of images is not effectively measured by the F1 accuracy as overall image classification tasks, as some side cases where the predicted mask does a poor job overlapping with the ground truth mask but the overall f1-score reflect satisfactory numbers, side cases like these prove the insuitability of using the F1-score for measuring the performance of semantic segmenation. Since we greatly care about maximizing overlaps between the ground-truth masks

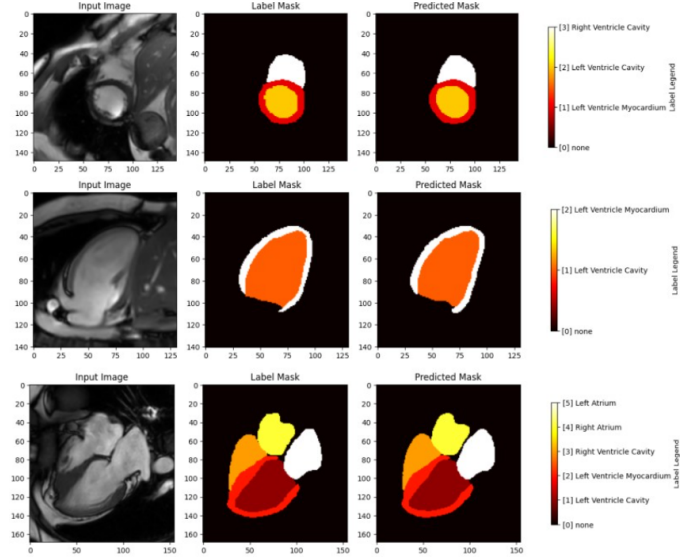


Fig. 16. A predicted mask sample of using the FCN model for inference on all views.

and out predicted masks, another metric dictates our measurement which measures this overlap effectively. Dice and intersection over union -IoU- scores (sometimes referred to as Jaccard Index) are utilized as our measurement metrics along with using plain f1 score normally. Formula for computing the Dice score and the IoU scores are shown by:

$$\text{Dice Score}(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|}$$

$$\text{IoU}(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}$$

Dice score is used to emphasize the importance of the mask more than background pixels. It is found that Dice score is mostly used in medical MRI segmentation as stated in [29]. However we are going to record the IoU and also the F1-score of our approaches.

B. Evaluating the FCN

The fully convolutional network (FCN) was used according to the setup in section III-A. The network is trained for 75 epochs using a Cosine Annealing learning rate scheduler with starting $\text{lr}=1\text{e-}4$ till $\text{lr}=1\text{e-}6$. For the model to focus on the core tissue rather than the background, a loss weight of 0.3 was assigned to the background pixels while 1.0 loss weight was assigned to all other classes regardless of their number (6 classes in 4CH problem, 4 classes in 2CH problem and 4 classes in SAX problem).

There is a noticeable jump of training and validation accuracies (and dice scores) when the learning rate scheduler was used rather than using a fixed learning rate.

This model is tested on the test set comprising of 30% of the original dataset length and the results were already promising and shown in table II. The model is used for inference for a final human check and a sample is shown in figure 16

TABLE II
PERFORMANCE COMPARISON OF THE FCN ACROSS DIFFERENT VIEWS

Metric	4CH	2CH	SAX
Dice Score	0.9325	0.9188	0.9423
Accuracy	0.9786	0.9819	0.9899
F1 Score	0.9424	0.9439	0.9557

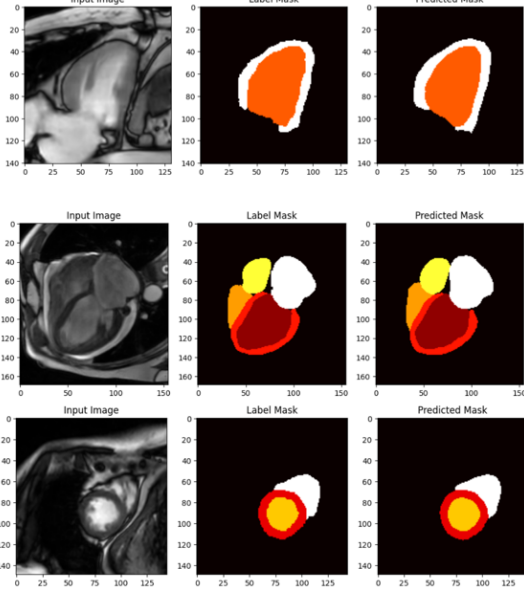


Fig. 17. Inference on real test cases using the proposed EG-FCN

C. Evaluating the EG-FCN

ORACM algorithm was implemented and was run offline on the training set to create another training set. Corresponding contour images and their original images were concatenated and input to the FCN for training for the same setup of 75 epochs with a learning rate cosine annealing scheduler. Results showed a push to the overall segmentation but the training time did not get improved, this architecture just elevated the roof of the model. The results are shown in table III and a sample of inference is shown in figure 17

TABLE III
PERFORMANCE COMPARISON ACROSS DIFFERENT VIEWS USING EG-FCN

Metric	4CH	2CH	SAX
Dice Score	0.9385	0.9229	0.9490
Accuracy	0.9806	0.9827	0.9910
F1 Score	0.9475	0.9467	0.9609

D. Evaluating the U-Net Architecture

For all experiments, the dataset was divided into training, validation, and testing sets to ensure reliable and unbiased evaluation of the segmentation model. The same preprocessing pipeline and experimental configuration were also applied to

the 2CH and SAX datasets to maintain consistency throughout the evaluation process.

The following subsections present the experimental results obtained from the 4CH, 2CH, and SAX cardiac MRI datasets.

1) *Evaluation on 4CH Cardiac MRI Dataset:* The model was trained for 10 epochs using the Adam optimizer together with the Cross-Entropy loss function.

To better analyze the learning behavior of the network during training, selected epochs were used to summarize the segmentation performance of the model. Table IV presents the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

The obtained results demonstrate that the proposed U-Net model achieved stable learning performance throughout the training process.

Figure 18 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 18, the model exhibited smooth convergence behavior during training without significant fluctuations or instability. The close similarity between the validation and testing results also indicates that the proposed framework achieved good generalization capability without severe over-fitting.

To further evaluate the segmentation quality across different cardiac structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table V summarizes the obtained segmentation performance on the testing dataset.

The per-class evaluation results demonstrate that the proposed segmentation framework was able to accurately identify multiple cardiac anatomical structures within the 4CH MRI images. Most segmentation classes achieved high precision and recall values, reflecting accurate feature extraction and reliable anatomical boundary localization.

Among all segmented structures, the LV cavity and left atrium achieved particularly strong segmentation performance with F1-scores of 0.96 and 0.95, respectively. Although the LV myocardium class achieved slightly lower performance compared with the remaining classes, the obtained results are still considered satisfactory because myocardial boundaries are relatively thin and more difficult to segment accurately in cine cardiac MRI images.

Figure 19 presents additional visualization examples generated by the proposed U-Net architecture on the 4CH cardiac MRI dataset.

The additional visualization results further confirm the robustness of the proposed segmentation framework and demonstrate the capability of the model to preserve fine cardiac anatomical details across different MRI samples.

2) *Evaluation on 2CH Cardiac MRI Dataset:* In the experimental evaluation of the proposed U-Net architecture on the 2CH cardiac cine MRI dataset, the model was trained for 10 epochs using the same preprocessing pipeline, optimization settings, and training configuration applied in the previous experiments.

TABLE IV
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE 4CH DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.4582	0.2255	0.2218	0.8312	0.9181	0.9188	0.8312	0.9181	0.9188
5	0.0696	0.0685	0.0686	0.9714	0.9717	0.9717	0.9714	0.9717	0.9717
10	0.0531	0.0551	0.0554	0.9777	0.9771	0.9770	0.9777	0.9771	0.9770

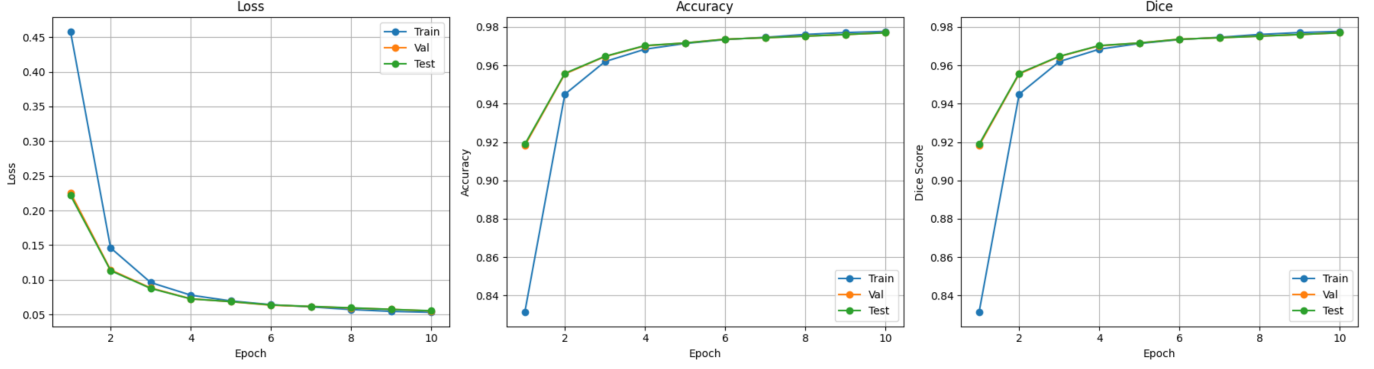


Fig. 18. Training performance curves of the proposed U-Net model on the 4CH cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during the training process.

TABLE V
PER-CLASS SEGMENTATION PERFORMANCE ON THE 4CH DATASET

Class	Precision	Recall	F1-Score
Background	0.99	0.99	0.99
LV Cavity	0.95	0.96	0.96
LV Myocardium	0.87	0.89	0.88
RV Cavity	0.94	0.93	0.93
Right Atrium	0.94	0.92	0.93
Left Atrium	0.94	0.95	0.95
Macro Average	0.94	0.94	0.94
Weighted Average	0.98	0.98	0.98

stable convergence behavior during the training process. The Dice score and segmentation accuracy increased consistently across the epochs, while the segmentation loss continuously decreased. Furthermore, the close agreement between validation and testing performance indicates good generalization capability without significant overfitting.

To further evaluate segmentation quality across different cardiac structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table VII summarizes the obtained per-class segmentation performance on the testing dataset.

Figure 21 presents qualitative segmentation examples generated by the proposed U-Net model on the 2CH dataset. The figure compares the original MRI slices, corresponding ground-truth masks, and the predicted segmentation outputs generated by the model.

3) *Evaluation on SAX Cardiac MRI Dataset:* The model was trained for 10 epochs using the same preprocessing pipeline, optimization settings, and experimental configuration applied in the previous experiments, for the SAX data.

To summarize the learning behavior of the proposed U-Net framework during training, selected epochs were used for performance comparison. Table VIII presents the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

Figure 22 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 22, the proposed model demonstrated highly stable convergence behavior during training. The Dice score and segmentation accuracy increased steadily across the

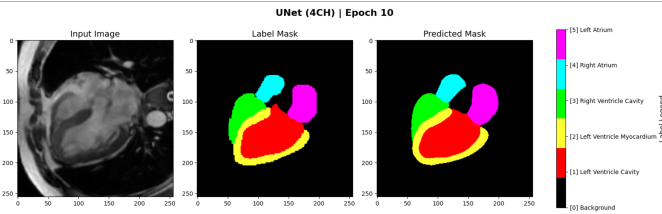


Fig. 19. Additional qualitative visualization results obtained from the proposed U-Net model on the 4CH cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

To summarize the learning behavior of the proposed U-Net framework during training, selected epochs were used for performance comparison. Table VI presents the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

Figure 20 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 20, the proposed model exhibited

TABLE VI
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE 2CH DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.3018	0.1730	0.1685	0.8770	0.9315	0.9336	0.8770	0.9315	0.9336
5	0.0598	0.0609	0.0598	0.9753	0.9747	0.9751	0.9753	0.9747	0.9751
10	0.0426	0.0488	0.0483	0.9820	0.9797	0.9799	0.9820	0.9797	0.9799

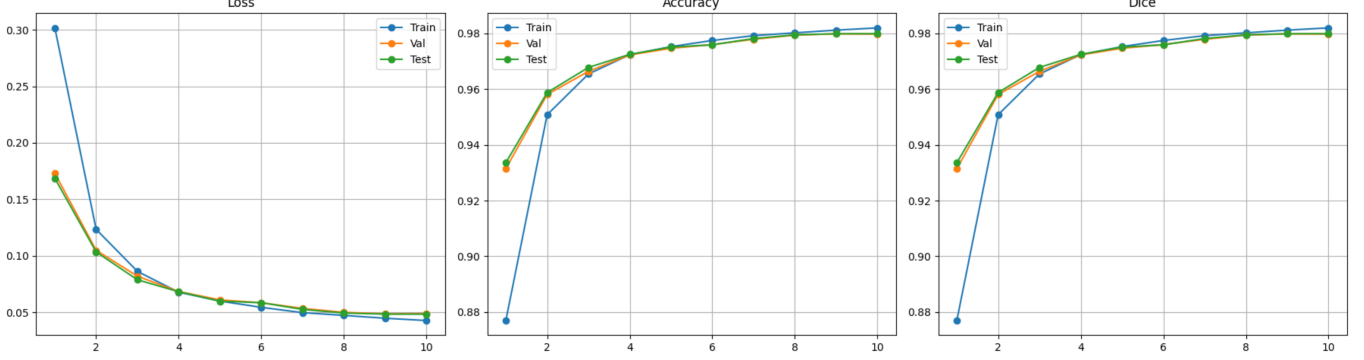


Fig. 20. Training performance curves of the proposed U-Net model on the 2CH cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during training.

TABLE VII
PER-CLASS SEGMENTATION PERFORMANCE ON THE 2CH DATASET

Class	Precision	Recall	F1-Score
Background	0.99	0.99	0.99
Left Ventricle Cavity	0.96	0.92	0.94
Left Ventricle Myocardium	0.87	0.90	0.89
Macro Average	0.94	0.94	0.94
Weighted Average	0.98	0.98	0.98

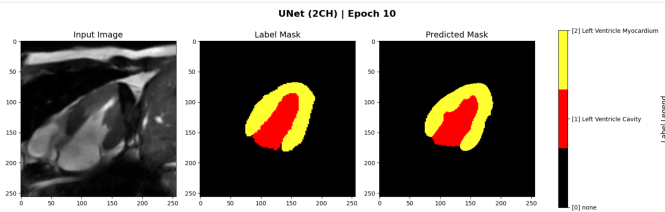


Fig. 21. Qualitative visualization results obtained from the proposed U-Net model on the 2CH cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

epochs, while the segmentation loss continuously decreased. In addition, the close similarity between validation and testing performance indicates strong generalization capability and reliable segmentation performance on unseen MRI images.

To further evaluate segmentation quality across different anatomical structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table IX summarizes the obtained per-class segmentation performance on the testing dataset.

Figure 23 presents qualitative visualization examples generated by the proposed U-Net architecture on the SAX cardiac MRI dataset.

E. Evaluating the Recurrent Residual U-net (R2U-net)

This proposed model was trained for 20 epochs with LR cosine annealing scheduler. Surprisingly, the model complexity greatly exceeded the dataset complexity and overfitted on the background class while misclassifying other classes. Results for the used Recurrent Residual U-net (R2U-net) is shown in table X. The learning curve of this model proves the heavy complexity of the model and is shown in figure 24 which indicates the need for further epochs and more training data.

F. Evaluating the U-Conv-LSTM

As explained in section III the update upon the U-net to make it make use of the time sequence feature between subsequent frames, this model was implemented and trained for 100 epochs with LR cosine annealing scheduler. Results were promising and did not show signs of overfitting, the metrics are stated in table XI. And a sample of inference is shown in figure 25

All utilized models are compared against each other as in table XII

VII. CONCLUSION

In this work, after performing exploratory data analysis on the dataset, we concluded that NIfTI data is best served with local Min-Max scaling without clipping. Our work utilized multiple Deep Learning models, – FCN, EG-FCN, U-net, R2U-net and U-Conv-LSTM –. Evaluating each model while tuning parameters and settings to achieve satisfactory results. We proposed jointing ORACM with FCN, implementing R2U-net for its increased complexity and U-Conv-LSTM network to make use of the time sequence feature. Results show that using ORACM achieves a slight push in segmentation while the

TABLE VIII
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE SAX DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.1641	0.0731	0.0724	0.9419	0.9725	0.9728	0.9419	0.9725	0.9728
5	0.0311	0.0323	0.0320	0.9873	0.9869	0.9870	0.9873	0.9869	0.9870
10	0.0221	0.0245	0.0249	0.9908	0.9901	0.9900	0.9908	0.9901	0.9900

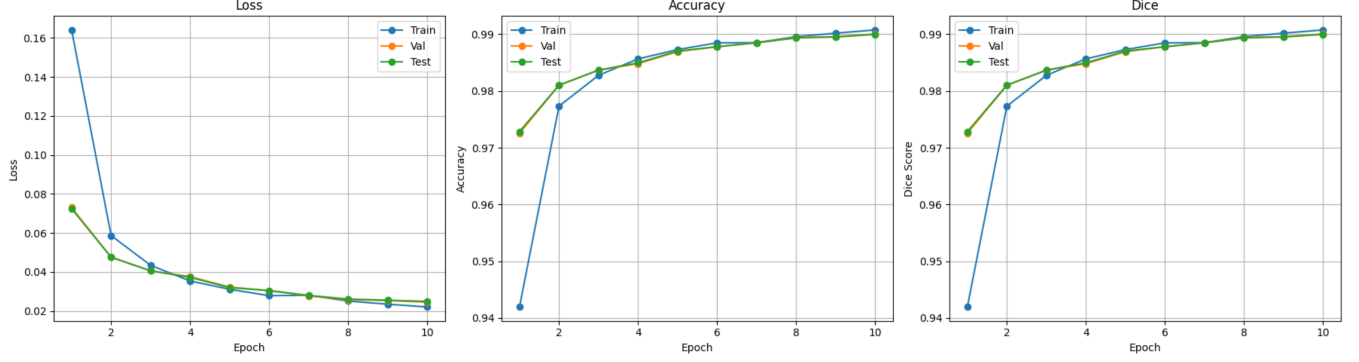


Fig. 22. Training performance curves of the proposed U-Net model on the SAX cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during the training process.

TABLE IX
PER-CLASS SEGMENTATION PERFORMANCE ON THE SAX DATASET

Class	Precision	Recall	F1-Score
Background	1.00	1.00	1.00
Left Ventricle Myocardium	0.93	0.94	0.93
Left Ventricle Cavity	0.98	0.95	0.96
Right Ventricle Cavity	0.95	0.94	0.95
Macro Average	0.96	0.96	0.96
Weighted Average	0.99	0.99	0.99

TABLE X
CLASSIFICATION REPORT FOR 4CH USING R2U-NET, INDICATING HEAVY OVERFITTING

Class	Precision	Recall	F1-score
BG	0.77	1.00	0.87
LV	1.00	0.00	0.00
MYO	0.00	0.00	0.00
RV	0.65	0.00	0.00
RA	0.00	0.00	0.00
LA	0.00	0.00	0.00
Macro Avg	0.40	0.17	0.15
Weighted Avg	0.69	0.77	0.67

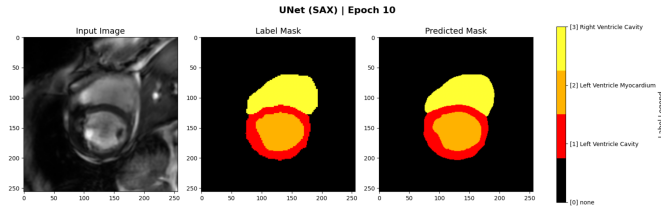


Fig. 23. Qualitative visualization results obtained from the proposed U-Net model on the SAX cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

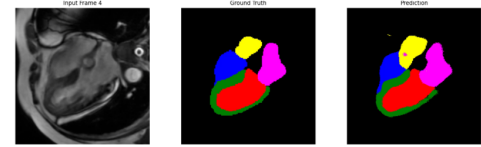


Fig. 25. Sample 4CH image using U-Conv-LSTM

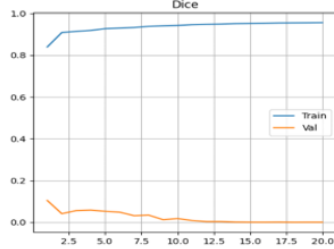


Fig. 24. Learning curve of the R2U-net, indicating heavy overfitting due to the model complexity

TABLE XI
TEST PERFORMANCE COMPARISON ACROSS DIFFERENT VIEWS USING U-CONV-LSTM

Metric	4CH	2CH	SAX
Test Loss	0.6972	0.6634	0.5042
Dice Score	0.8188	0.8414	0.7669
Accuracy	0.9346	0.9426	0.9417

TABLE XII
PERFORMANCE COMPARISON OF DIFFERENT SEGMENTATION MODELS

Model	Dice Score	Accuracy	F1 Score
CNN with Interpolation	0.3100	0.2900	0.2100
FCN	0.9325	0.9786	0.9424
EG-FCN	0.9385	0.9806	0.9475
U-Net	0.9911	0.9905	0.9900
R2U-Net	0.0050	0.7700	0.6700
U-Conv-LSTM	0.8188	0.9346	0.8000

bare metal U-net is unmatched in-comparison with its hybrid models (R2U-net and U-Conv-LSTM).

VIII. FUTURE WORK AND OPENNESS FOR IMPROVEMENTS

There is an unvisited problem in the CINE CMR images, which is the misalignment of slices due to the motion of the patient. MRIs are slow and the patient usually moves due to continuous breathing, which can cause misalignment between slices. This misalignment can cause imperfections during training or even introduce distorted frames. To address this issue, each frame should have a reference point and adjacent frames should undergo reference mapping. Another suggestion for improvement is the association of information from different views together for the same case. If a patient has a record of 2CH, 4CH and SAX views CMR, we can make use of the fact that all those image views belong to the same patient and try to correlate those views together for a more accurate classification or prediction or even use those different views for depth estimation. Generative AI can also be used to construct a projected 3D image view of the heart given different CMR views with missing slices where the Generative AI model will predict and estimate those missing slices and construct a 3D instance of the patient's heart.

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